

PURCHASE DESCRIPTION

COAT, ALL-WEATHER, MEN'S, ARMY

This purchase description is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE. This purchase description covers the requirements for a men's double-breasted all-weather coat with a removable liner worn by male military personnel of the Department of the Army. The all-weather coat is a modified commercial item, since a specific military appearance is required (see 3.1).

2. CLASSIFICATION. The coat will be of one (1) type in the following sizes:

2.1 Schedule of sizes.

Size	X-Short	Short	Regular	Long	X-Long
30	--	X	X	--	--
32	X	X	X	X	--
34	X	X	X	X	X
36	X	X	X	X	X
38	X	X	X	X	X
40	X	X	X	X	X
42	X	X	X	X	X
44	--	X	X	X	X
46	--	X	X	X	X
48	--	X	X	X	X
50	--	X	X	X	--
52	--	X	X	X	--
54	--	X	X	X	--

Comments, suggestions, or questions on this document should be addressed to: Department of the Army, Combat Capabilities Development Command (CCDC) Soldier Center, 10 General Greene Avenue, Natick MA 01760, ATTN: FCDD-SCP-WC.

AMSC N/A

FSC 8405

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3. SALIENT CHARACTERISTICS.

3.1 Description. The coat (see figures 1 and 2) shall have a double-breasted front with a six (6) button closure, a convertible collar with button closure, a belt with buckle, a two-piece back with a center vent with a single button closure, a back yoke, split raglan sleeves, button-down pointed shoulder straps, button-down pointed sleeve straps, a 1/2 front and a 1/2 back lining with fully lined sleeves, and a removable insulated liner (see figures 3, and 4). The material for the sleeve lining shall be taffeta as specified in 3.3.2.2. The coat front shall have two (2) vertical welt pockets with pass through slits, a 3/8-inch wide hanger loop at back neck with a minimum finished opening of 1-7/8-inches, and extra buttons sewn on right front facing. The coat shall have a right front flap (gun patch).

3.1.1 Belt and buckle. The belt shall finish 2-1/8 ($\pm$ 1/8)-inches wide and shall have a buttoned belt anchor on the left side with a belt loop keeper, fused interlining, and a buckle. The buckle shall be a slide type buckle and conform to the dimensions specified in figure 5. The color of the buckle shall be a good match to the basic material (see 7.7.1).

3.1.2 Belt loop placement. The coat shall have four (4) belt loops, finishing 2-1/2 ($\pm$ 1/8)-inches in length and 3/8 to 1/2-inch wide. The belt loops shall be securely bartacked at the top and bottom of belt loop. The belt loops shall be positioned as follows:

One (1) belt loop on each front, approximately 1/4-inch above and in line with front pocket seam line.

One (1) belt loop on each side seam positioned in horizontal alignment line with the front belt loops.

The belt loop on the left side seam shall have an under belt loop for the belt anchor.

3.1.3 Shoulder straps. The finished shoulder straps shall be centered along the shoulder seam and the pointed end shall finish even to 1/2-inch from the collar setting seam. The shoulder straps shall finish smooth and flat without distortion. The overedged straight end of shoulder strap shall be attached to the coat with a 3/16-inch seam, then shoulder strap shall be folded to the finished position and shall be top stitched 1/4-inch from the folded edge. The shoulder strap topstitching shall follow the top stitching line of the front sleeve seams.

3.1.4 Sleeve straps. The sleeve straps shall be positioned on the back seam of the sleeve in accordance with notches on the pattern. The pointed end of the sleeve strap shall finish even to 1/2-inch from sleeve top seam. The lower edge of the sleeve strap shall be parallel to the bottom edge of sleeve. The sleeve straps shall finish smooth and flat without distortion.

3.1.5 Label placement. The coat shall have a combination identification/instruction/size label. The combination label shall be stitched on all four sides to the bottom right facing. The combination label shall not be visible when coat is worn. The removable liner shall have a size label caught in the neck or side seam.

3.1.6 Removable liner. The removable liner shall have bound edges, 1/2 raglan sleeves, two (2) bound pass-through pocket slits and a center back bound vent. The liner's sleeve bottom shall be bound and shall have binding forming a button loop's opening on the underarm sleeve seam. The

button loop shall accommodate a 24 ligne button (see figures 3 and 4).

3.1.6.1 Slide fastener. The slide fastener for attaching the liner to the coat shall be a separating, non-reversible right hand, light-size with an aluminum or brass chain and a short tab pull. The color of the tape shall be a good match to the basic material and shall be 7/16 or 9/16-inch wide. The ends of the slide fastener shall be equidistant from the bottom edge of the coat.

3.2 Seams and stitches. All joining and topstitch seams shall have 10 to 12 stitches per inch. All safety stitch, serging or basting seams shall contain 8 to 10 stitches per inch. The width or bight of the overedge stitching shall be not less than 3/16-inch. All bartacks shall have a minimum of 28 stitches per tack and shall be a minimum of 3/8-inch in length. The following components shall be bartacked or line tacked: One (1) end of belt anchor to belt, belt keeper to inside of belt, top and bottom edges of belt loops to coat. The back lining shall be tacked to coat at the side seams. Raise stitching and edge stitching shall be 1/16-inch from the edge. Safety stitch shall be 3/8-inch gage. All seams shall be sewn so that no raw edges, run offs, twists, pleats, puckers, or open seams shall result. All seams shall start and finish evenly. Thread tension shall be maintained so that the stitching will not be tight or loose and the stitching shall be balanced. The ends of all seams not caught in other seams or stitching shall be backtacked not less than 1/2-inch.

3.2.1 Topstitching. The following components shall be finished with two (2) rows of topstitching 1/16-inch from the edge with 1/4-inch gage utilizing lock stitching: shoulder straps, sleeve straps, collar, pocket welt, belt, top and front edges of the lapel, front edges of the coat and bottom of front facing, outside back yoke hem and gun patch(s). The side seams and top sleeve seams shall be safety stitched and two-needle topstitched or two-needle felled, 1/16-inch from the edge with 1/4-inch gage. The front and back sleeve seam shall have a single row of 1/4-inch topstitching extending to step as marked on pattern. The left side of the back vent and the back seam shall have a single row of lockstitch 1/4-inch from edge. The front and back lining shall have a 3/8 to 5/8-inch clean finished hem.

3.2.2 Buttonholes. The front and lapel buttonholes shall have a 1-1/4-inch cut opening, and shall be horizontal eyelet end square-bar type, worked over gimp, with a minimum of 70 stitches (including bartack). The horizontal buttonholes shall be positioned in accordance with the pattern marker, three (3) on the left front, one (1) on the right front in horizontal alignment with top button on right front, one (1) on each lapel, and the cut opening of buttonholes shall be positioned 3/4-inch from edge of coat. Each shoulder strap and sleeve strap shall have a 7/8-inch eyelet end, square-bar type worked over gimp buttonhole positioned with cut opening centered, $\pm$ 1/8-inch on strap and 7/8-inch from pointed end of strap. There shall be two (2) 3/4-inch cut, square-bar type buttonholes, one (1) centered on the belt fastener 3/8-inch from the end and one (1) centered on the right back vent as per mark on pattern. The lapel buttonholes shall be 3/4-inch from the edge and 1-1/2-inches down from the top.

3.2.3 Buttons. The coat shall have eight (8) 45 ligne buttons; three (3) buttons on each front, one (1) button on the inside left front with a shank wrapped 45 ligne anchor button to properly engage buttonhole on the right front, and one (1) on the upper right front to correspond with the left lapel buttonhole. The right front buttons shall be set on a stand and attached to an 18 ligne stay button on the inside and then shank wrapped. The left front buttons shall be attached to 18

ligne stay buttons and shall be sewn with a tight stitch, they shall not be shank wrapped. The coat shall have four (4) 30 ligne buttons; one (1) for each sleeve to interface with the sleeve strap buttonhole and one (1) on each shoulder to interface with the shoulder strap buttonhole. The coat shall have four (4) 24 ligne buttons; one (1) button shall be sewn on each sleeve lining to correspond with the loops on the removable liner sleeve's bottom, one (1) shall be positioned on the belt to correspond to belt anchor buttonhole and one (1) on the left back vent to correspond to the buttonhole. Extra buttons, one (1) of each size (45, 30, 24, and 18) shall be sewn on the lower inside right facing.

3.2.3.1 Lapel button placement. The top right lapel button shall be position so that the left coat front lays smooth and flat without distortion and the neck opening shall be 2 to 2-1/2-inches between collar stand joining seam and collar stand joining seam across front of coat.

3.3 Materials.

3.3.1 Basic material. The basic material for the outer shell and body lining of the coat shall be water-repellent treated polyester/cotton poplin cloth. The cloth shall be a plain weave, and shall match the standard sample Heritage Green 564 for shade, appearance, and colorfastness. The finished cloth shall conform to the physical requirements and tests specified in Table I. The cloth shall match the standard sample when viewed using the AATCC Evaluation Procedure 9, Option C (see 7.9), with sources simulating artificial daylight D75 illuminant with a color temperature of 7500K(± 200) illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (± 200). The use of dyes or substances containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. The basic material shall be free of liable sulfur when tested as specified in 3.3.1.1.

TABLE I. Basic material requirements.

Characteristic	Requirement	Test Method
Fiber content, ± 3 percent: Polyester Cotton	65 percent 35 percent	AATCC 20 AATCC 20
Weight. oz./sq. yd.	5.3 to 6.8	ASTM D3776/D3776M
Yarn ply: Warp Filling	Single or 2 ply Singles	Visual Visual
Yarns per inch (min.): Warp Filling	108 68	ASTM D3775 ASTM D3775
Weave	Plain	Visual

TABLE I. Basic material requirements. - Continued

Characteristic	Requirement	Test Method
Air permeability (max.)	10	ASTM D737
Breaking strength (min.): Warp Filling	185 lbs. 70 lbs.	ASTM D5034 ASTM D5034
Tearing strength (min.): Warp Filling	5.0 lbs. 2.0 lbs.	ASTM D1424 ASTM D1424
Water repellency: spray test (min.): Initial After 1 dry-cleaning	100, 100, 90 70, 70, 70	AATCC 22 AATCC 158 & AATCC 22
Water repellency, tumble jar (max.): Dynamic absorption test Initial	18 percent 18 percent	AATCC 70 AATCC 70
Hydrostatic pressure (min.): Initial After 3 launderings After 1 dry-cleaning	30 25 25	AATCC 127 AATCC 127 & AATCC 96, VIc, A AATCC 127 & AATCC 158
Resistance to organic liquid (N-Tetradecane) (min.): Initial (min.) After 3 launderings	No wetting No wetting	AATCC 118 AATCC 96, VIc, A & AATCC 118
Dimensional stability, percent (max.): Warp Filling	2 2	AATCC 96, VIc, A AATCC 96, VIc, A
Colorfastness to (min.): Laundering, after 3 cycles Perspiration (acid and alkaline) Crocking (wet and dry) Light (after 40 AFU or 170 kJ/(m ² nm)@420 nm) <u>2</u> /	3 4 3 4	AATCC 61, 2A AATCC 15 <u>1</u> / AATCC 8 AATCC 16.3
pH	5.0 - 8.5	AATCC 81

1/ Except that the pH shall be 12.7 (+0.2).

2/ (AFU) AATCC Fading Units

3.3.1.1 Presence of labile sulfur. In the determination of labile sulfur in textile materials with lead acetate, two (2) 1.50 gram (± 0.01) gram samples from each material submitted for evaluation shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. The samples shall be submersed in a stannous chloride solution that contains 100 grams of stannous chloride crystals ACS in 100 millimeters of hydrochloric acid ACS (35 percent concentration) and 50 milliliters of distilled water. A filter paper wet out with a 5.0 percent lead acetate solution shall be placed over the top of the test tube. The lead

acetate solution contains 5.0 grams of lead acetate CP reagent grade and enough distilled water to make up a 100 milliliter solution; if the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The test tube containing the textile sample, stannous chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows.

- | | |
|----------|--|
| Free | - The filter paper shows no discoloration of staining of any kind. |
| Slight | - The filter paper shows a light tan to light brown discoloration stain. |
| Moderate | - The filter paper shows a dark brown discoloration stain. |
| Severe | - The filter paper shows a black color stain. |

3.3.2 Removable liner materials. The insulated removable liner's material shall be constructed of a layer of batting sandwiched between a top layer and bottom layer of taffeta material. The three layers shall be channel quilted together. The channels shall be a minimum of $5 \pm 1/2$ -inches wide and the stitching shall be 301 stitch type with 11 stitches per inch.

3.3.2.1 Batting. The batting material for the body of the removable insulated liner shall be a thin, lightweight, double scrim batting composed of 65 percent polyolefin/35 percent polyester. The batting for the body of the liner shall weigh a minimum of 4.6 ounces per square yard (weight excludes scrim) and shall be a minimum of 0.4-inch thick.

3.3.2.2 Taffeta. The taffeta shall consist of continuous filament 70 denier yarn, with a minimum of 100 yarns per inch in the warp direction and 68 yarns per inch in the filling direction. The taffeta shall weigh $1.5 (\pm 0.15)$ ounces per square yard when tested in accordance with ASTM D3775, and ASTM D3776/D3776M, Option C, respectively. The color shall be a good match to the basic material and shall have colorfastness requirements equal to that of the basic material when tested in accordance with the test methods listed for colorfastness in Table I.

3.3.2.3 Binding material. The bias binding shall be cut from the taffeta material specified in 3.3.2.2 and the color shall be a good match to the basic material and have colorfastness requirements equal to that of the basic material. Commercial binding is acceptable if the shade and physical properties of the taffeta are met.

3.3.2.4 Removable liner's sleeve material. The material for the removable liner's sleeves shall be a multi-filament acetate faced, combed cotton or staple rayon backed, 5 or 8 harness satin weave material weighing a minimum of 6.2 ounces per square yard when tested in accordance with ASTM D3776/D3776M. The material shall have a minimum of 194 ends per inch and a minimum of 46 picks per inch when tested in accordance with ASTM D3775. The material shall have a minimum warp breaking strength of 76 pounds and a minimum filling breaking strength of 75 pounds when tested in accordance with ASTM D5034. The material shall be finished so as to provide a lustrous face and a napped back. The material for the removable liner's sleeves shall show "good" colorfastness to gas fading, perspiration; and wet-drycleaning, and shall have a minimum rating of 3.5 to dry crocking and a minimum rating of 3 to wet crocking when tested according to AATCC 164, AATCC 15, AATCC 132, and AATCC 8, respectively, using the

AATCC Chromatic Transfer Scale. The material shall not shrink or elongate more than 4 percent in the warp direction nor more than 2 percent in the filling direction when tested as specified in AATCC 96, Test VIc, A. The color of the finished cloth (face side) shall be a good match to the basic material.

3.3.3 Hanger loop material. The hanger loop shall be made from the basic material and shall be 3/8-inch wide.

3.3.4 Fusible interlining. The coats interlining shall be a nonwoven fusible material consisting of 85 percent polyester and 15 percent polyamide or 100 percent polyester, with a weight of 1.75 to 2.0 ounces per square yard, with a polyamide dot adhesive. The color shall be charcoal or black, and shall have a colorfastness to dry-cleaning (3 cycles) minimum rating of 3.0 when tested in accordance with AATCC 132. For optimum fusing results, specific fuse times and settings should be obtained from the manufacturer of the non-woven fusible.

3.3.5 Threads. The thread for seaming and stitching shall be cotton-covered, polyester core, 36 - 45 Tex, 2 ply thread for needle, bobbin, or looper or 31 - 35 Tex, 2 ply thread for bobbin or looper. The thread for button sewing shall be cotton-covered, polyester core thread, 61 - 70 Tex, 2 or 3 ply. The thread shall conform to A-A-50199. The threads' finish shall be non-staining and non-flame propagating. The threads shall be treated with a permanent water repellent and the color shall be a good match to the basic material. The threads shall have a rating of 3-4 for colorfastness to wet dry cleaning when tested in accordance with AATCC 132.

3.3.6 Gimp. The gimp for reinforcing buttonholes shall be cotton conforming to Size No. 8, Type I or II of A-A-50198. The gimp shall have a rating of 3-4 for colorfastness to dry cleaning when tested as specified in A-A-50198. The color shall be a good match to the basic material.

3.3.7 Buckle. The plastic buckle (See figure 5) for the belt shall be made from a phenol-formaldehyde, cellulose filled molding material, having a minimum Izod impact strength of 0.46 foot-pounds per inch of notch when tested in accordance with 3.3.7.2. The size shall be 2.250 inches with an overall outside width of 1.625 (± 0.2) inches and length of 2.875 (± 0.2) inches, inner slot. The color shall be a good match to the basic material. All corners shall be rounded and have a buckle thickness of 0.421 (± 0.02) inches (reference figure 5). The front surface of all buckles shall be slightly curved with a pebbled surface to simulate leather and the outer rim shall be beaded to simulate stitching. The finished buckles shall be free of flash and the outside surface shall be produced by molding. The application of a finish or protective coating of lacquer or other finishing materials shall not be permitted. Knockout marks are permitted on the back of the buckles provided such marks are no deeper than 0.015-inch. The buckle shall have a minimum breaking strength of 100 pounds for lot sample average and 90 pounds for individual sample when tested as specified in 3.3.7.1. The buckle shall have a minimum impact strength of 7 inch-pounds for lot sample average and 5 inch-pounds for individual sample without evidence of cracking or breaking when tested as specified in 3.3.7.2.

3.3.7.1 Breaking strength. Breaking strength shall be determined on a pendulum type tensile machine. The pulling grip shall have a speed at no load of 12 ($\pm 1/2$)-inches per minute. Attachment of the buckle to the grips shall be accomplished by loops of webbing or other strap

material passing around the side bars of the buckle with ends of the strips centered in the grips of the testing machine. Width of the material shall be 2-inches minimum and 2-3/16-inches maximum. Distance between grips at strut of test shall be 6-inches. Specimens shall be tested to failure with three (3) tests per sample to the nearest pound. Breaking strength tests may be run under room temperature conditions for pre-production and inspection tests. However, in the case of referee tests, all specimens to be tested shall be conditioned at standard laboratory conditions, 73.5 ($\pm$ 2)°F and 50 ($\pm$ 4) percent relative humidity for a minimum of 40 hours.

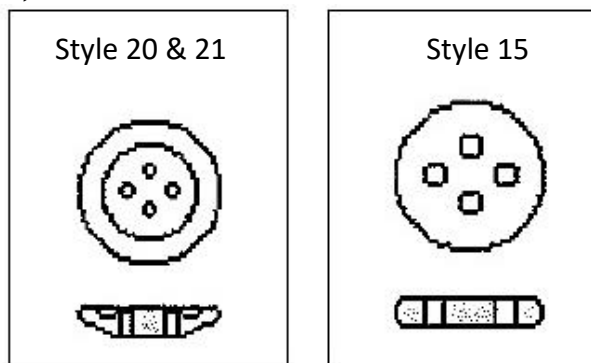
3.3.7.2 Impact strength. Impact strength shall be tested in accordance with ASTM D256 with three (3) tests per sample to the nearest 0.1 inch-pound. Impact tests may be run under room temperature conditions for pre-production and inspection tests. However, in the case of referee tests, all specimens to be tested shall be conditioned at standard laboratory conditions, 73.5 ($\pm$ 2)°F and 50 ($\pm$ 4) percent relative humidity for a minimum of 40 hours.

3.3.8 Buttons. The buttons shall be sewing hole buttons of a melamine formaldehyde composition conforming to the styles indicated in Table II. The buttons shall have a glossy finish and the color shall be a good match to the basic material. The buttons shall not exhibit chalking when tested as specified in 3.3.8.2 and shall conform to the characteristics specified in Table II.

TABLE II. Buttons.

Button Placement	Style <u>1/</u>	Size Diameter Ligne	Compression strength, lbs., min. <u>2/</u>	Thickness Ligne, min. <u>3/</u>	Total quantity per coat
Front & inside anchor	20 or 21	45	800	7	8
Sleeve & shoulder strap	20 or 21	30	800	5	4
Sleeve liner, belt, back vent	20 or 21	24	800	5	4
Stay buttons	15	18	800	4	6
Extra buttons	20 or 21	45	800	7	1
	20 or 21	30	800	5	1
	20 or 21	24	800	5	1
	15	18	800	4	1

1/ Description of styles 20, 21 and 15.



2/ When tested in accordance with 3.3.8.1

3/ Thickness shall be reported to the nearest full ligne.

3.3.8.1 Compressive strength test. The buttons shall meet the minimum requirement when tested for compressive strength using an apparatus, which permits gradual application of the load either, by a manual or automatic hydraulic mechanism. Buttons shall be placed face down one at a time between flat blocks of steel, tested to failure. Failure is defined as the first sign of a crack in the button visible to the naked eye (a visible crack in the button will usually be found at the first audible sound of cracking). The load in pounds, which produces this failure, shall be recorded as the compressive strength of the button.

3.3.8.2 Chalking test. Buttons shall be immersed in a boiling solution of 0.8 percent by weight sulfuric acid for 10 minutes and immediately thereafter dried and examined by holding the buttons at arm's length and examining them under a strong light. Any chalking as evidenced by a dry chalk-like appearance or deposit on the surface of the button shall be considered a test failure.

3.3.9 Slide fastener. The slide fastener for the coat and removable liner shall be a separating, non-reversible, right-hand, light size with an aluminum or brass chain and short tab pull. The lengths of the slide fasteners shall conform to the size coat, with a minimum length of 70-inches. The color of the tape shall be a good match to the basic material and shall be 7/16 to 9/16-inch wide. The ends of metal mechanism of the slide fastener shall be equidistant from the bottom hem of coat shell and straight bottom hem of the liner, $\pm 1/2$ -inch.

3.3.10 Seam tape. The seam tape for the topstitched seams shall be a lightweight, thermally bonded open net structure formed from pure polyamide or polyester extruded adhesive and shall weigh a minimum of 0.5 ounces per square yard and a maximum of 1.0 ounces per square yard. The polyamide adhesive shall be nylon 6, 6,6, 11, 12 or any co-polymer blends thereof and shall be applied to the substrate by powder dot, paste dot, hot melt dot, bi-component dot, sintered, or spun fused process. The polyester adhesive shall be polyester resin applied to the substrate by powder dot, paste dot, hot melt dot, bi-component dot, sintered, or spun fused process. The color of the seam tape shall be natural. For optimum fusing results, specific fuse times and settings should be obtained from the manufacturer of the fusible seam tape.

3.4 Labels.

3.4.1 Coat's combination identification/instruction/size label. The combination identification/instruction/size label shall be white, commercial type, sewn in label (see 3.1.5). The label's inscription shall be in bold face type and shall have a minimum font size of 10 points, and shall have a rating of 3-4 for colorfastness to laundering when tested in accordance with AATCC 61-3A. The label shall be sewn down on all four sides and shall be positioned on the right front facing, as worn, with the bottom of label 9 (± 1)-inches from bottom of finished hem. The stitching shall not cover the printing and shall not be visible on the front of the coat. The inscription's legibility, the label, and label attachment shall last the expected life of the coat. The label shall include, but not be limited to, the following information:

Coat, All-Weather, Men's 65% Polyester/ 35% Cotton (Example) SPO100-00-0-000 (Example) XYZ Manufacturing (Example) Dry Clean Only After each dry cleaning, professionally retreat coat for water repellency 36R (Example) NSN: 8410-00-000-0000 (Example)

3.4.2 Liner's size label. Each removable liner shall have a white, commercial type size label (see 3.1.5). The label's inscription shall have a minimum font size of 10 points, and shall have a rating of 3-4 for colorfastness to laundering when tested as specified in AATCC 61-3A. The label shall be placed in the side seam. The inscription's legibility, the label, and label attachment shall last the expected life of the coat. The label shall be in bold face type and include the following information:

36R (Example)

NOTE: The length (X-Short, Short, Regular, Long and X-Long) may be abbreviated as follows: XS, S, R, L, or XL.

3.4.3 Barcode label/tag. Each coat shall have a bar code label/tag in accordance with Type VIII, Class 17 of MIL-DTL-32075. The bar coding element shall be a 13 digit National Stock Number (NSN). There shall be a 12 digit Universal Product Code (UPC) number assigned for all NSNs by the Government. The initial "UPC" must appear beneath the code. The bar codes for NSN and UPC type shall be a medium to high density and shall be located so that it is completely visible on the coat when it is folded and/or packaged as specified. The barcodes shall be clearly legible and readable by a scanner. The label's location shall cause no damage to the coat.

3.4.4 Certification label. Each coat, when produced for Army & Air Force Exchange Services (AAFES) or commercial sales, shall have a label attached stating certification information as described in 5.7 (see 7.8). The certification label shall be positioned on the right front facing, 1/2-inch below the combination identification/instruction/size label and stitched on all four sides. The stitching shall not cover the printing and shall not be visible on the front of the coat. As an alternate, the certification label may be combined with the combination label.

3.4.4.1 Age code labeling. Each certification label shall have an age code for quarter and year certification was issued placed at the bottom of the certification label. The age code may be accomplished by direct printing, stamping or stenciling.

3.5 Patterns. Standard patterns, to be used for cutting the contractor's working patterns, shall be furnished by the Government. Minor pattern modifications are permitted to accommodate automation and manufacturing procedures; however, the design and finished measurements

must be maintained. The patterns provide for a seam allowance of 1/4-inch for shoulder strap, collar, sleeve straps, belt, and front facing. The patterns provide a seam allowance of 3/8-inch for safety stitched seams and 1/2-inch seam allowance for double lapped, double stitched seams.

3.5.1 Pattern version. PD19_026 dated 13 March 2020.

3.5.2 List of pattern parts. The pattern list in Table III is provided to insure that the pattern set is complete.

TABLE III. List of pattern parts.

Material	Nomenclature	Pattern Abbreviation	Cut Parts
Basic material (see 3.3.1)	Back	PD19_026-BACK	2
	Front	PD19_026-FRONT	2
	Front sleeve	PD19_026-SLV-FR	2
	Back sleeve	PD19_026-SLV-BK	2
	Front facing	PD19_026-FRONT-FAC	2
	Belt	PD19_026-BELT	1
	Collar	PD19_026-CLLR	2
	Collar stand	PD19_026-CLLR-STND	2
	Pocket facing	PD19_026-PCKT-FACING	2
	Sleeve strap	PD19_026-SLV-STRP	2
	Shoulder strap	PD19_026-SHLDR-STRP	4
	Back sleeve cap reinforcement	PD19_026-SLV-BK-CAP-REINFR	2
	Pocket welt	PD19_026-PCKT-WLT	2
	Gun flap right	PD19_026-GUN-FLP-RGT	2
	Back yoke	PD19_026-BACK-YOKE	1
	Under pocket	PD19_026-PCKT-UNDR	2
	Top hanging pocket	PD19_026-PCKT-HANG-TOP	2
	Neck facing	PD19_026-NECK-FAC	1
	Back half lining	PD19_026-BACK-LINING	1
	Front half lining	PD19_026-FRONT-LINING	2
Lining (see 3.3.2.2)	Front sleeve lining - taffeta	PD19_026-SLV-FR-LIN-TFFT	2
	Back sleeve lining - taffeta	PD19_026-SLV-BK-LIN-TFFT	2
Fusible interlining (see 3.3.4)	Front facing interlining	PD19_026-FRONT-FAC-INT	2
	Belt interlining	PD19_026-BELT-INT	1
	Shoulder strap interlining	PD19_026-SHLDR-STRP-INT	2
	Pocket welt interlining	PD19_026-PCKT-WLT-INT	2
	Collar interlining	PD19_026-CLLR-INT	2
	Collar stand interlining	PD19_026-CLLR-STND-INT	2
	Sleeve strap interlining	PD19_026-SLV-STRP-INT	2
	Front sleeve interlining	PD19_026-SLV-FR-INT	2
	Back sleeve interlining	PD19_026-SLV-BK-INT	2
	Gun flap right interlining	PD19_026-GUN-FLP-RGT-INT	1
	Neck facing interlining	PD19_026-NECK-FAC-INT	1

TABLE III. List of pattern parts. - Continued

Material	Nomenclature	Pattern Abbreviation	Cut Parts
Liner, Removable Sateen (see 3.3.2.4)	Liner front sleeve - sateen	PD19_026-LINER-SLV-FR	2
	Liner back sleeve - sateen	PD19_026-LINER-SLV-BK	2
Liner, Removable, Quilted (see 3.3.2.1)	Liner back - quilted	PD19_026-LINER-QUILTED-BK	1
	Liner front - quilted	PD19_026-LINER-QUILTED-FR	2
Marker	Button marker	PD19_026-BUTTON-MARKER	0

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3.6 Finished measurements. The finished coats and belts shall conform to the measurements in Table IV, which are expressed in inches:

TABLE IV. Coat finished measurements.

SIZE	Tol.	30	32	34	36	38	40	42	44	46	48	50	52	54
Chest 1/														
All lengths	±3/4	39	41	43	45	47	49	51	53	55	57	59	61	63
Back Length 2/														
X-Short	±1/2	--	40-3/4	41	41-1/4	41-1/2	41-3/4	42	--	--	--	--	--	--
Short	±1/2	42-1/2	42-3/4	43	43-1/4	43-1/2	43-3/4	44	44-1/4	44-1/2	44-3/4	45	45-1/4	45-1/2
Regular	±1/2	44-1/2	44-3/4	45	45-1/4	45-1/2	45-3/4	46	46-1/4	46-1/2	46-3/4	47	47-1/4	47-1/2
Long	±1/2	--	46-3/4	47	47-1/4	47-1/2	47-3/4	48	48-1/4	48-1/2	48-3/4	49	49-1/4	49-1/2
X-Long	±1/2	--	--	49-1/2	49-3/4	50	50-1/4	50-1/2	50-3/4	51	51-1/4	--	--	--
Sleeve Length 3/														
X-Short	±1/2	--	16	16	16	16	16	16	--	--	--	--	--	--
Short		17	17	17	17	17	17	17	17	17	17	17	17	17
Regular	±1/2	18	18	18	18	18	18	18	18	18	18	18	18	18
Long	±1/2	--	19	19	19	19	19	19	19	19	19	19	19	19
X-Long	±1/2	--	--	20-1/2	20-1/2	20-1/2	20-1/2	20-1/2	20-1/2	20-1/2	20-1/2	--	--	--
Belt Length 4/														
All lengths	±3/4	40	42	44	46	48	50	52	54	56	58	60	62	64

The coat shall be placed flat on a table and measured as specified below:

- 1/ Chest: Twice the measurement taken with coat buttoned from folded edge to folded in line with pit of armhole.
- 2/ Back length: The measurement taken from center back seam from lower edge of under collar to bottom edge of coat.
- 3/ Sleeve length: The measurement taken along the underarm seams from the base of armhole to bottom end of sleeve.
- 4/ Belt length: The measurement taken, with the belt in a flat position, along the center of belt from center of buckle to pointed end of belt.

3.6.1 Liner measurements. The liner shall conform to the measurements in Table V, expressed in inches:

TABLE V. Liner finished measurements.

SIZE	Tol.	30	32	34	36	38	40	42	44	46	48	50	52	54
Chest 1/														
All lengths	±3/4	31	33	35	37	39	41	43	45	47	49	51	53	55
Back Length 2/														
X-Short	±1/2	--	33	33-1/4	33-1/2	33-3/4	34	34-1/4	--	--	--	--	--	--
Short	±1/2	34-3/4	35	35-1/4	35-1/2	35-3/4	36	36-1/4	36-1/2	36-3/4	37	37-1/4	37-1/2	37-3/4
Regular	±1/2	36-3/4	37	37-1/4	37-1/2	37-3/4	38	38-1/4	38-1/2	38-3/4	39	39-1/4	39-1/2	39-3/4
Long	±1/2	--	39	39-1/4	39-1/2	39-3/4	40	40-1/4	40-1/2	40-3/4	41	41-1/4	41-1/2	41-3/4
X-Long	±1/2	--	--	41-3/4	42	42-1/4	42-1/2	42-3/4	43	43-1/4	43-1/2	--	--	--
Sleeve Length 3/														
X-Short	±1/2	--	14-1/2	14-1/2	14-1/2	14-1/2	14-1/2	14-1/2	--	--	--	--	--	--
Short	±1/2	15-1/2	15-1/2	15-1/2	15-1/2	15-1/2	15-1/2	15-1/2	15-1/2	15-1/2	15-1/2	15-1/2	15-1/2	15-1/2
Regular	±1/2	16-1/2	16-1/2	16-1/2	16-1/2	16-1/2	16-1/2	16-1/2	16-1/2	16-1/2	16-1/2	16-1/2	16-1/2	16-1/2
Long	±1/2	--	17-1/2	17-1/2	17-1/2	17-1/2	17-1/2	17-1/2	17-1/2	17-1/2	17-1/2	17-1/2	17-1/2	17-1/2
X-Long	±1/2	--	--	18-1/2	18-1/2	18-1/2	18-1/2	18-1/2	18-1/2	18-1/2	18-1/2	--	--	--

The liner shall be placed flat on a table and measured as specified below:

1/ Chest: Twice the measurement taken with liner folded in half (down center back) and measured from front bound edge to back folded edge in line with pit of armhole and multiplied by two.

2/ Back length: The measurement taken from center back neck edge to bottom of liner.

3/ Sleeve length: The measurement taken along the underarm seams from pit of armhole to bottom of sleeve.

3.7 Toxicity. The cloth shall be given a water repellent treatment and the cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 5.6. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.8 Workmanship. The finished coats and liners shall conform to the quality of product established by this purchase description. The occurrence of defects shall not exceed the applicable acceptable quality limits (see 7.6).

4. REGULATORY REQUIREMENTS. Unless otherwise specified the offer/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. Each coat and removable liner provided shall meet the salient characteristics of this purchase description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same or similar product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Market acceptance. The coat and liner must have been sold to the commercial market or the Government.

5.3 End item inspection procedures. Sampling for inspection shall be conducted in accordance with ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection of Attributes (see 7.2.3).

5.4 Visual examination. The coat and liner shall be examined for the defects listed in Table VI. The lot size shall be expressed in units of coats. The sample unit is one (1) coat. The inspection level and AQL shall be as specified in the contract or purchase order. Unless otherwise specified, scoring shall be performed on defects visible from a normal inspection distance of 3-feet.

TABLE VI. Visual examination.

Examination	Defect
Material and workmanship	Hole, cut, tear, smash, misweave, missing, broken, or loose yarn, shade bar, dye streak, area of no dye penetration Any component part full, tight, omitted, improper, twisted or not cut as specified Any component part not cut in accordance with specified pattern part Measurement of components not as specified, finished measurements not as specified Fronts uneven in length at top or bottom by more than 1/4-inch; front facing exposed on outside of front for a length of 1-inch or more Open or puckered seam, seams and stitching not as specified, needle chew, irregular stitching, any component part caught in unrelated stitching, any seam twisted or pleated, any stitching omitted or insecure, or a raw edge Bartack, tack, or tacking omitted, insecure, misplaced, not serving intended purpose, or not as specified Visible mend, pleat or crease Needle perforations clearly visible on outside Irregular stitching or stitching not as specified, or not specified stitches per inch Fusible omitted, any evidence of fusible bleed-through, bubbling, delamination or color change; fusible exposed Seam tape omitted Any buttonhole with broken stitches, ragged edges, misplaced, omitted The edge of any component part required to be forced out having folds of more than 1/8-inch
Collar and lapel	Collar or lapel not uniform in size or shape (more than 1/8-inch difference in symmetry between left and right edges), collar twisted, not smooth and flat, short or tight causing collar or lapel to turn out
Front flaps	Front flaps omitted, not positioned as specified, not specified size or shape
Vent	Vent opening reversed, vent sides uneven at bottom by more than 1/4-inch, not constructed as specified
Hanger loop	Hanger loop omitted, improperly positioned, or insecure
Sleeves	Sleeves set improperly
Pockets	Pockets out of alignment by more than 1/2-inch, pocket welt not uniform in size and shape, pocket opening not smooth and flat, pocket notch exposed on outside, pass through slits omitted or not properly positioned
Epaulets and sleeve tab	Epaulet or sleeve tab not uniform in size or shape or does not lay flat, sleeve tab uneven distance from bottom of sleeve by more than 1/2-inch, buttonhole not centered on epaulet or sleeve tab

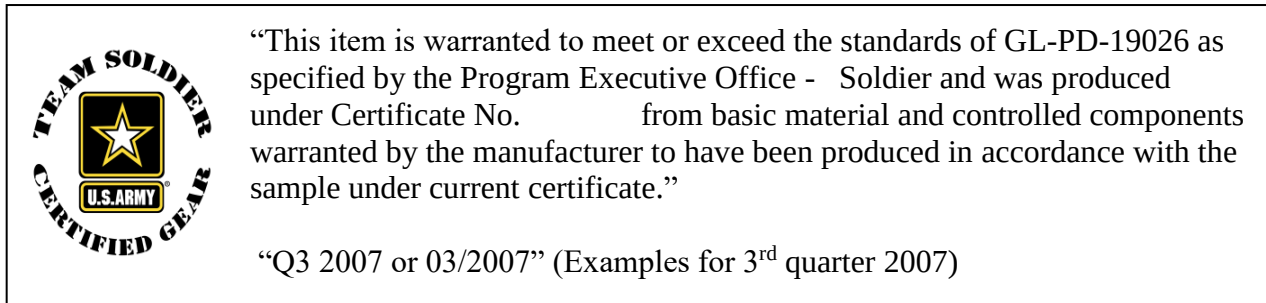
TABLE VI. Visual examination. - Continued

Examination	Defect
Buckles	Buckle omitted, defective or not as specified
Buttons	Any button misplaced, omitted, broken or defective, not attached as specified, button not aligned with corresponding buttonhole causing gap, puckers, or tightness, any part of coat puckered or pleated by button sewing
Slide fastener	Slide fastener not functioning properly, pull detaches, slide binds or sticks, chain unmeshed, coat or liner twisted or distorted when slide fastener is engaged, slide fastener tape stitched to close to chain not permitting slide fastener to move freely. Slide fastener wavy or not lying smooth and flat, Slider positioned on right side of liner
Binding	Binding omitted, twisted, or not securely caught on fabric, End of binding on liner sleeve bottom not finishing at underarm seam, or not formed into a loop to accommodate a 24-ligne button
Pressing	Pressing omitted, wrinkles, creases, or pressing impressions not removed, scorched area
Color	Shaded parts on the outside, any part noticeably shaded from the other parts or variation in shade within a part Thread, button or buckle color not as specified
Cleanliness	Any spot or stain clearly noticeable, five (5) or more thread ends on outside not trimmed to 3/8-inch or less or not removed; odor affecting serviceability
Labels	Omitted, incorrect, illegible, not attached where specified; barcodes omitted, not readable by scanner; human-readable interpretation (HRI) omitted or illegible; barcode not visible on folded, packaged item; bar code attachment causes damage to the item
Packaging	Any coats not packaged in accordance with the contract or purchase order

5.5 End item dimensional examination. The coats and liners shall be examined for conformance to the finished measurements specified in Tables IV and V. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of coats. The sample unit shall be one (1) coat. The inspection level, expressed in terms of defects per hundred units, shall be as specified in the contract or purchase order.

5.6 Toxicity test. When required (see 7.6), an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of the studies indicate the coat is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 7.2.4). If the toxicity requirement (see 3.7) can be demonstrated with historical use data, on finishing treatments used, toxicity testing may not be required.

5.7 Certification. When the coats are purchased individually or in lots by AAFES or a commercial supplier, the supplier shall certify to the purchaser that the items have been made in accordance with the quality of material and components established by this purchase description and shall include a label incorporating certification number assigned. In addition, the label shall have an age code for quarter and year certification was issued and placed at the bottom of the certification label. The label shall read as follows:



5.8 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order (see 7.6).

6. PACKAGING

6.1 Packaging. The preservation, packing and marking shall be as specified in the contract or purchase order (see 7.6).

7. NOTES

7.1 Sources of Government documents.

7.1.1 Copies of Government documents are available online at <https://quicksearch.dla.mil/>.

7.1.2 Rules and Regulations under the Textile Fiber Products Identification Act are available online at <https://www.ftc.gov>.

7.2 Sources of non-Government documents.

7.2.1 AATCC test methods are available online at <https://www.aatcc.org>.

7.2.2 ASTM Standards are available online at <https://www.astm.org>.

7.2.3 ASQ/ANSI Z1.4 – Sampling Procedures and Tables for Inspection of Attributes are available online at <https://asq.org>.

7.2.4 INFORMA HEALTHCARE - Repeat Insult Patch Test – Modified Draize Procedure – Principles and Methods of Toxicology, A. Wallace Hayes (editor) are available online at <https://www.crcpress.com>.

7.3 Standard samples. For access to standard samples address the contracting activity issuing the invitation for bids or request for proposal.

7.4 Intended use. The all-weather coat is intended for wear by male personnel of the Army as part of the Army Green Service Uniform (AGSU).

7.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

7.6 Ordering data. The contract or purchase order should specify the following:

- a. Title, number, and date of this purchase description.
- b. Size required (see 2.1).
- c. Labeling information (see 3.4).
- d. When toxicity testing is required (see 3.7 and 5.6).
- e. Regulatory requirements (see 4).
- f. Product conformance provisions (see 5.1).
- g. Acceptance criteria (see 5.8).
- h. Packaging requirement (see 6.1).

7.7 Known sources.

7.7.1 Buttons and Buckle. US Button Shade M3772 has been found to meet the requirements for the button and the buckle.

7.7.2 Thread. A&E Colonial Brown has been found to meet the shade requirements for the thread.

7.8 Certification Procedures. For a copy of the US Army Uniform Quality Control Program (UQCP) Operating Procedures please contact Department of the Army, Combat Capabilities Development Command (CCDC) Soldier Center, 10 General Greene Avenue, Natick, MA 01760, ATTN: FCDD-SCP-WC.

7.9 Visual shade matching. In 2019, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be “The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 ($\pm$ 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens”.

7.10 Summary of changes since last revision.

- a. Page 1: Added size 54 to Schedule of Sizes and requirements throughout PD for this size
- b. Page 3, 3.2, changed stitches per inch to 8-10

- c. Page 5, Table I and 3.3.2.4, updated colorfastness to crocking requirements
- d. Page 11: Add pattern date
- e. Pages 11 and 12: Add new List of pattern parts
- f. Updated format and typographical errors throughout

7.11 Subject terms.

Army Green Service Uniform (AGSU)
Water repellent

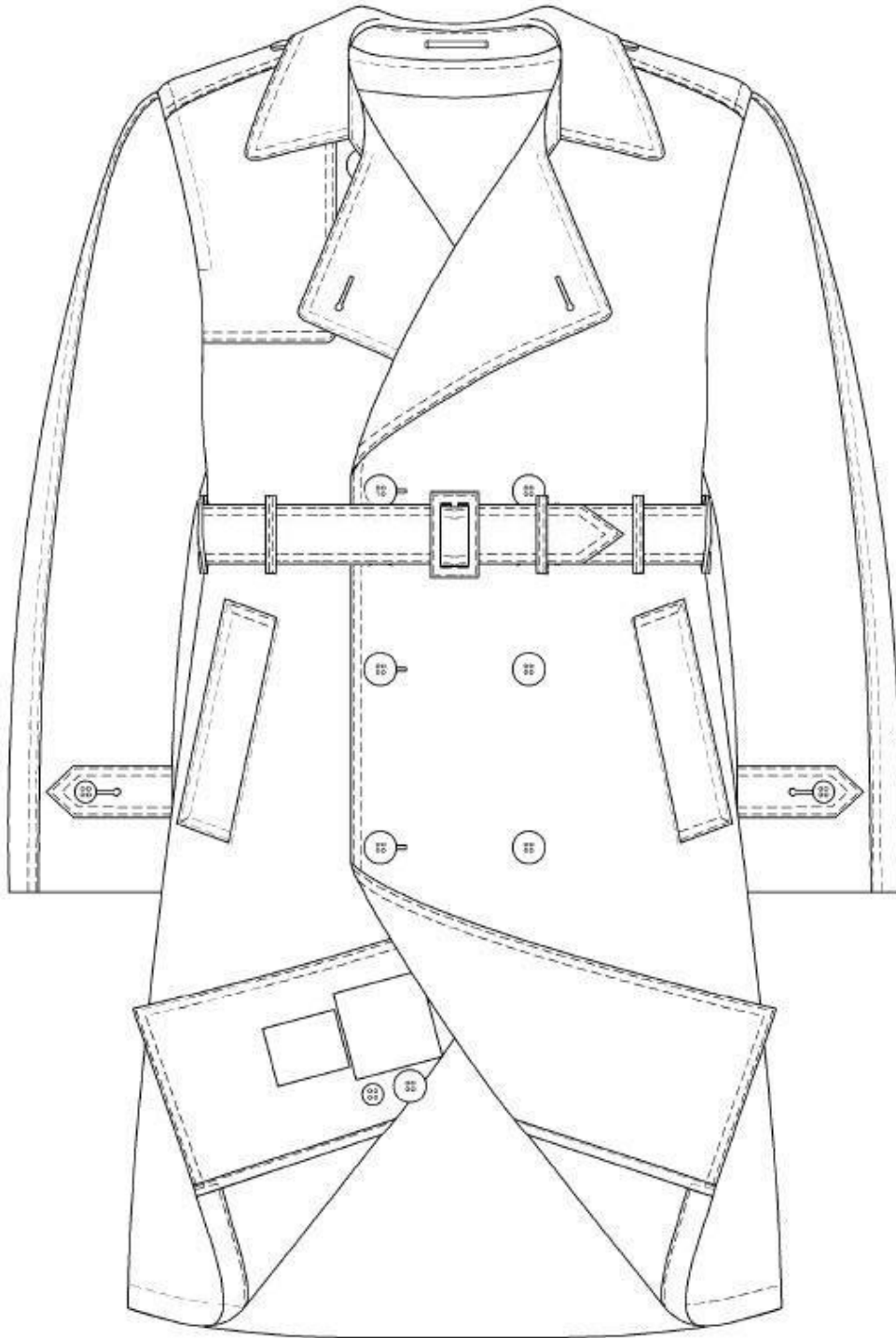


FIGURE 1. Coat, All Weather, Men's, Front

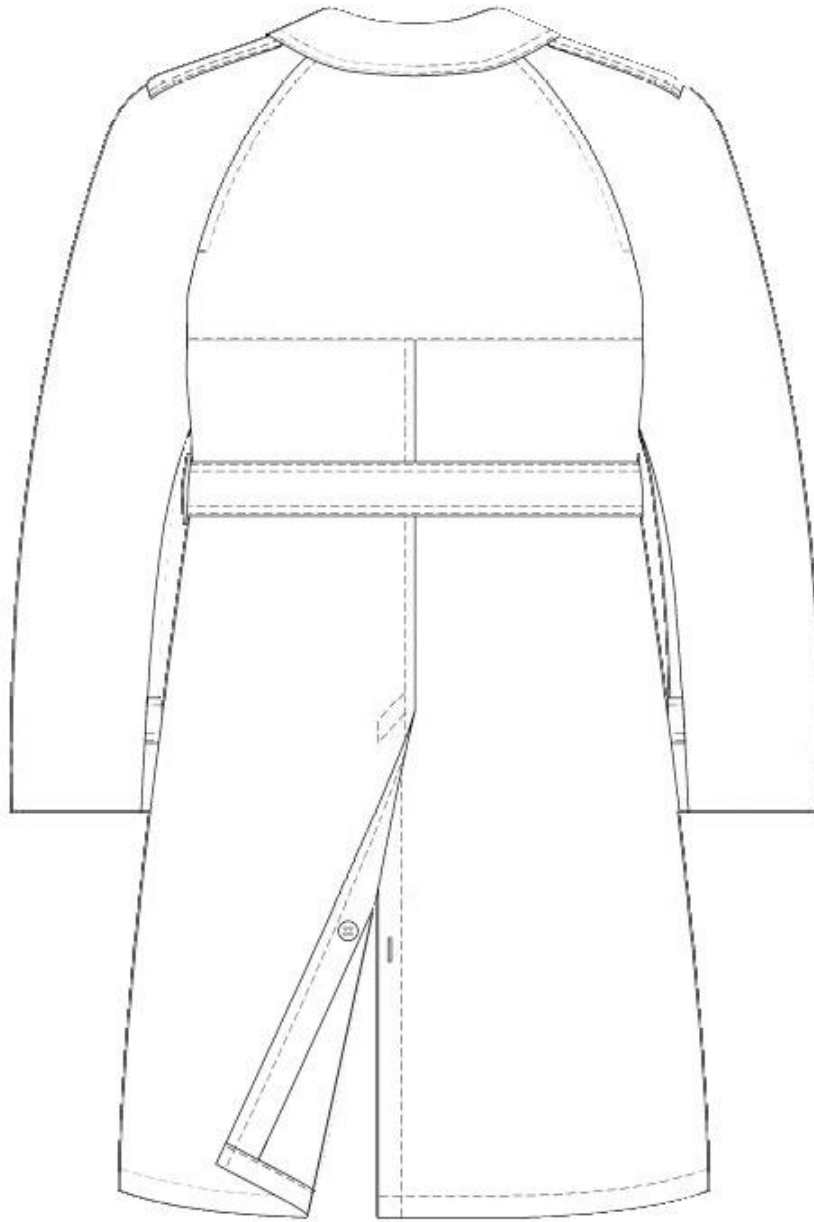


FIGURE 2. Coat, All Weather, Men's, Back

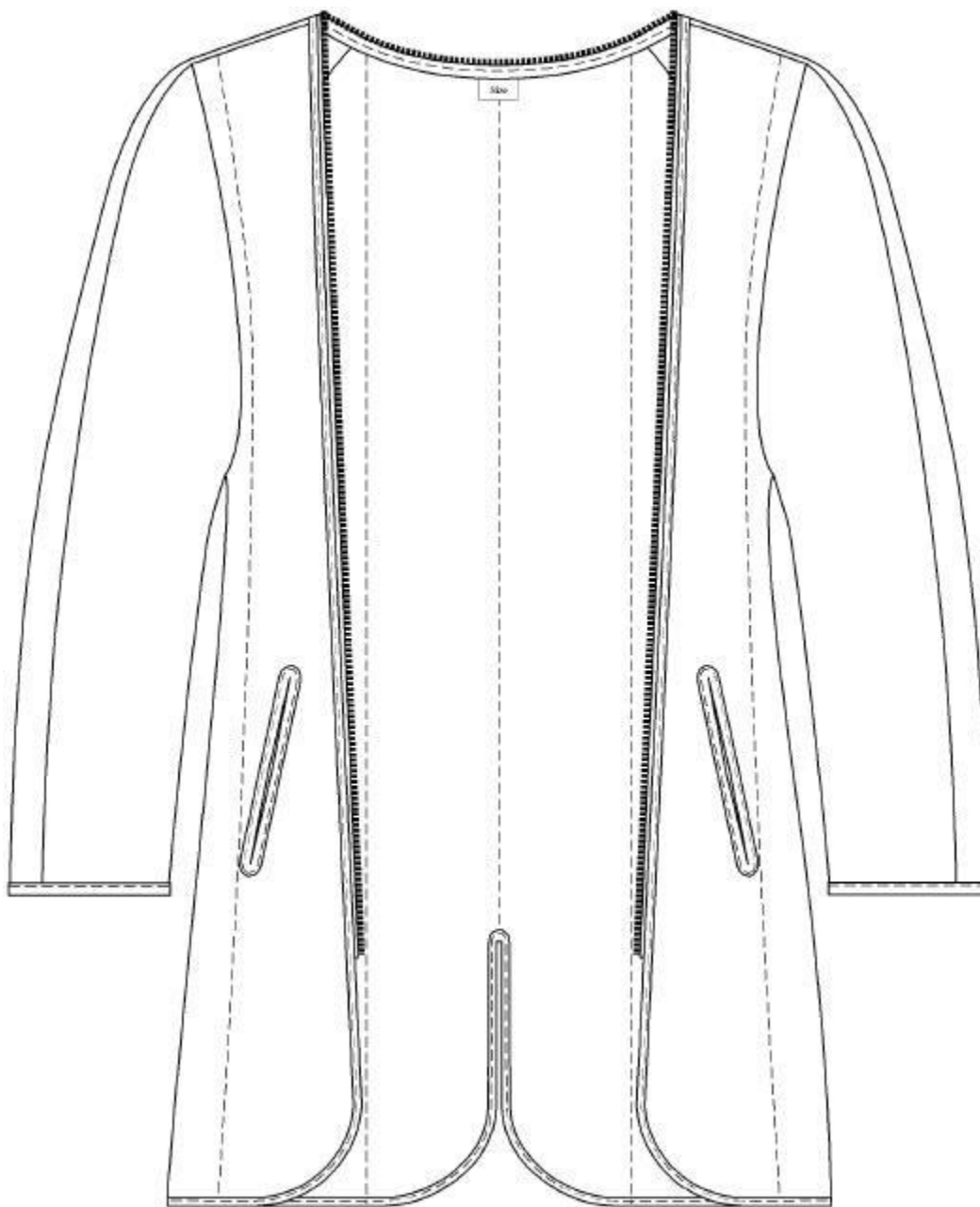


FIGURE 3. Removable Liner, Men's, Front

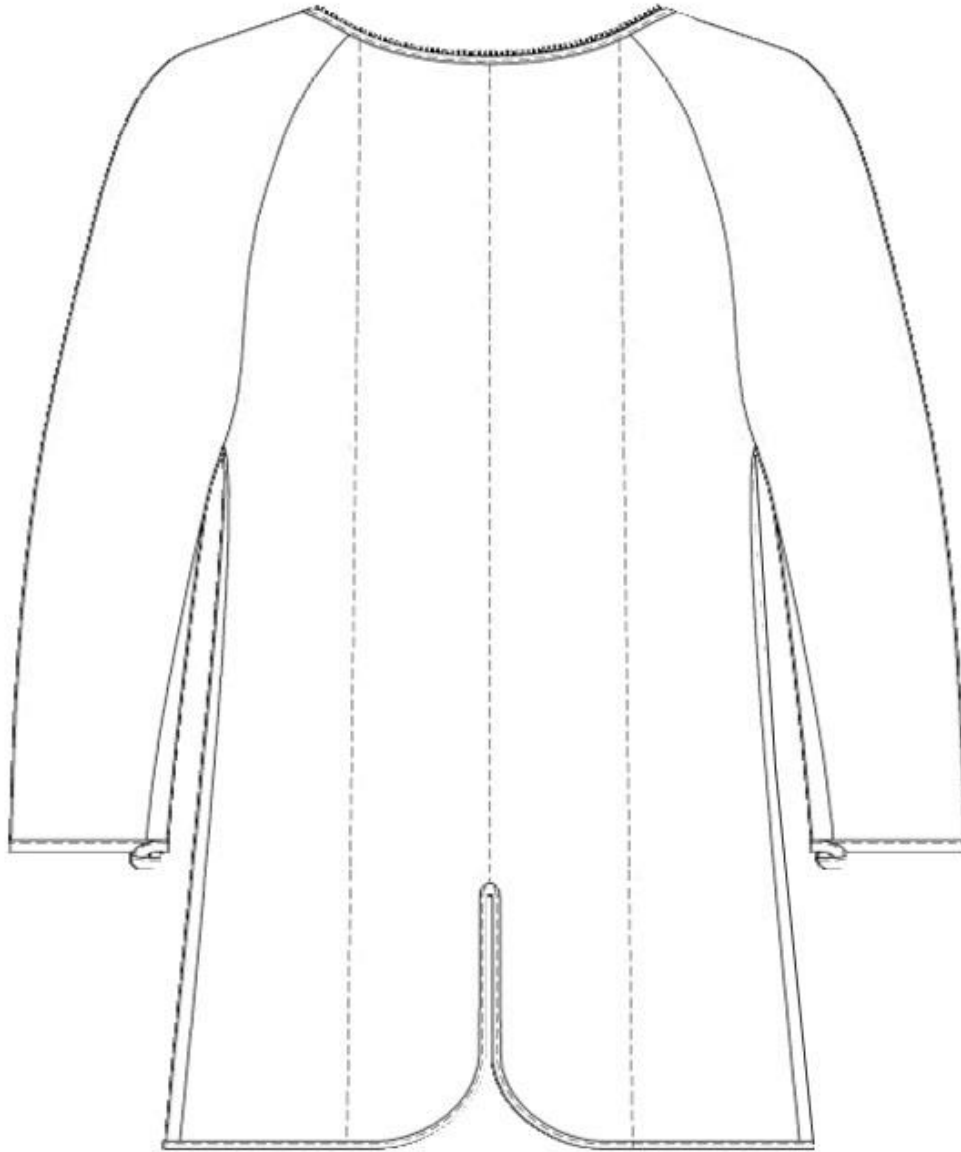


FIGURE 4. Removable Liner, Men's, Back

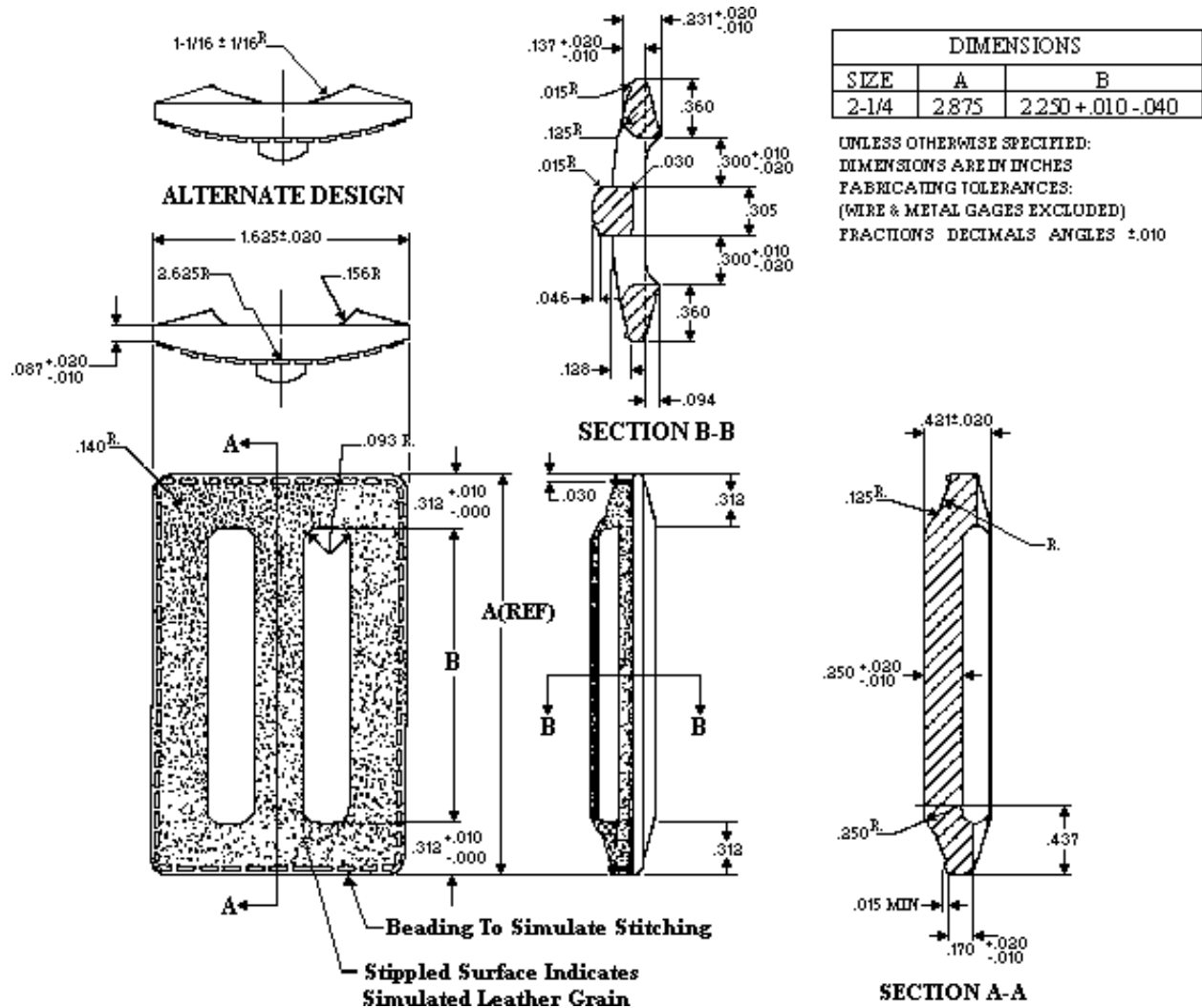


FIGURE 5. Belt Buckle

Custodian:
Army-GL

Preparing Activity:
Army – GL